Frontend: Theory Assignment 2

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1. What is CSS Box Model?

The CSS Box Model explains how every HTML element is treated as a box. It consists of four parts: content, padding, border and margin. Content is the actual text or image, padding is the space around the content, border surrounds the padding, and margin is the space outside the border.

2. What is media query? Explain with a sample code.

A media query is used in CSS to make a web page responsive for different screen sizes. It applies different styles when a condition, such as screen width, is true. Example: @media (max-width: 600px) { body { background-color: lightblue; } }

3. What are the features of Bootstrap?

Bootstrap is a popular front-end framework. Its main features are a responsive grid system, ready-made components, utility classes, buttons and forms, responsive design, and support for JavaScript components. It also helps in making websites faster and easier to design.

4. Write an example of basic grid structure in Bootstrap.

A basic Bootstrap grid is made using rows and columns. Example: Column 1 Column 2 Column 3 . The columns share the available width equally.

5. What is CSS Selectors?

CSS selectors are used to select HTML elements on which we want to apply CSS styles. Common selectors are element selector, class selector, id selector, universal selector and attribute selector. For example, p selects all paragraph elements, .box selects elements with class box, and #title selects the element with id title.

6. What are CSS backgrounds, list the properties?

CSS background properties are used to set the background of an element. Important properties are background-color, background-image, background-repeat, background-position, background-size, background-attachment and background. These properties can be used separately or together.

7. How can we overlap elements in CSS?

Elements can be overlapped by using the position property and z-index. For example, an element can be given position: absolute and placed using top, left, right or bottom. The z-index property controls which element appears in front of another element.

8. What are the different positioning properties in CSS?

The main CSS positioning values are static, relative, absolute, fixed and sticky. Static is the normal position. Relative moves an element from its normal position. Absolute positions an element relative to its positioned parent. Fixed keeps an element fixed on the screen. Sticky behaves like relative until a scrolling position is reached and then acts like fixed.

9. What is ECMAScript?

ECMAScript is a standard or specification for scripting languages. JavaScript is the most well-known implementation of ECMAScript. It defines features such as variables, functions, objects, classes and other language rules. New ECMAScript versions add improvements to the language.

10. What is the difference between var, let and const in JavaScript. Explain Temporal Dead Zone?

var, let and const are used to declare variables in JavaScript. var is function-scoped and can be redeclared and reassigned. let is block-scoped and can be reassigned but cannot be redeclared in the same block. const is block-scoped and cannot be reassigned after declaration. The Temporal Dead Zone, or TDZ, is the time between entering a block and the declaration of a let or const variable. During this time, accessing the variable causes a ReferenceError.